INCH-POUNDS

MIL-PRF-62198A(AT)
7 November 1995
SUPERSEDING
MIL-V-62198(AT)
31 May 1983

PERFORMANCE SPECIFICATION

VALVE, SELECTOR, HYDRAULIC DUMP

This specification is approved for use within U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the M60 tank selector valve assembly which allows the pressure of the hydraulic system to be dumped (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E, Warren, MI 48397-5000 by using the Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

AMSC N/A FSC 4820

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, and standards. The following specifications, and standards, form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-H-46170 - Hydraulic Fluid, Rust Inhibited, Fire-resistant, Synthetic Hydrocarbon Base.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810 - Environmental Test Methods.
MIL-STD-1246 - Product Cleanliness Levels and Contamination Control Program.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWINGS

DEPARTMENT OF DEFENSE

11674266 - Valve, Selector.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents that are adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the non-government documents that are current on the date of the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspections by Attributes.

(Application for copies should be addressed to The Society for Quality Control, P.O. Box 305, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first article sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for valve selector assemblies. The materials shall be capable of meeting all of the operational and environmental requirements specified herein. Recovered materials shall be used to the maximum extent practicable (see 4.6.1).

3.2.1 Material compatibility. Selector valve assembly materials shall be compatible with hydraulic fluid per MIL-H-46170 (see 4.6.1).

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3.2.2 Corrosion control. The selector valve assembly shall be fabricated from materials that are inherently corrosion resistant or are permanently treated to prevent forms of corrosion and deterioration which affect system performance (see 4.6.1).

3.3 Design and construction. Selector valve assembly shall conform to the manufacturer's design and construction except for the envelope dimensions; port hole dimensions, locations, and depth; lever swing radius and location; and mounting dimensions which shall conform to drawing 11674266 and other applicable drawings (see 4.6.2).

3.3.1 Interchangeability. All selector valve assemblies shall be interchangeable as to fit, form, and function with other manufacturer's assemblies having the same military part number (see 4.6.2).

3.3.2 Weight. The dry weight of the selector valve assembly shall not exceed 2 pounds (0.9 kilogram (kg)) (see 4.6.2).

3.4 Performance.

3.4.1 Torque. The torque required to move the handle out of the open or closed detented positions shall be 8 $\pm$ 6 inch pounds (0.9 $\pm$ 0.68 Newton-meter (N-m)) with 1000 pounds per square inch gage (psig) (6.88 Megapascal (MPa)) of hydraulic fluid applied to port 1 with port 2 blocked (see drawing 11674266). The torque required to move the handle between detent positions shall not exceed 15 inch pounds (1.7 N-m) (see 4.6.4).

3.4.2 Pressure drop. Pressure differential between port 1 and port 2 shall not exceed 750 psi (5.16 MPa) at a minimum flow of 5 gallons per minute (gpm) (18.9 liters per minute (L/min)) when the handle is in the open position. Pressure differential at environmental extremes shall not exceed 1500 psi (7.9 MPa) (see 4.6.5).

3.4.3 External leakage. There shall be no external leakage of the selector valve (see 4.6.6).

3.4.4 Internal leakage. With the valve handle in the closed position, the leakage from port 2 shall not exceed 0.003 cubic inches (in^3) (0.05 cubic centimeters (cm^3)) per minute at any pressure applied to port 1 up to 3000 psig (20.6 MPa). Leakage from port 2 shall not exceed 0.006 in^3 (0.1 cm^3) per minute after endurance performance (see 4.6.7).

3.4.5 Proof pressure. Selector valve, in the closed position, shall withstand a minimum pressure of 4500 psig (31 MPa) for a duration of 3 minutes without permanent deformation, rupture, or external leakage (see 4.6.8).

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3.4.6 Burst pressure. Selector valve, in the closed position, shall withstand a minimum of 75 psig (51.6 MPa) for a duration of 2 minutes without bursting. The valve need not be operable after application of this pressure (see 4.6.9).

3.4.7 Environmental conditions. Selector valve shall operate as specified herein under the temperature, shock and vibration conditions specified in 4.6.10.

3.4.7.1 High temperature. Selector valve shall operate as specified herein after exposure to ambient temperature of 160°Fahrenheit (°F) (71° Celsius (°C)) for 36 hours. Valve assembly shall operate during and after exposure to 125°F (51°C) (see 4.6.10.1).

3.4.7.2 Low temperature. Selector valve shall operate as specified herein after exposure to ambient temperature of -65°F (-54°C) for a minimum of 12 hours. Valve assembly shall operate during and after exposure to -25°F (-31°C) (see 4.6.10.2).

3.4.7.3 Shock resistance. Selector valve shall operate as specified herein, after exposure to a shock impulse of 40 gravity units (g) for 18 ± 3 milliseconds (ms), half sine (see 4.6.10.3).

3.4.7.4 Vibration resistance. Selector valve shall operate as specified during and after exposure to vibration levels profiled in figure 1 for an 80 minute period (including up to four resonance dwells of 13.5 minutes each) with each of three mutually perpendicular axes (see 4.6.10.4).

3.4.8 Endurance. Selector valve assembly shall operate as specified herein after being subjected to 10,000 actuation cycles with 1,000 psig (6.88 MPa) applied to port 1 while port 2 is blocked (see 4.6.11).

3.5 Internal cleanliness. Selector valve assembly shall meet the cleanliness requirements as specified in 4.6.12.

3.6 Identification and marking. The selector valve assembly shall include, as a minimum, the following information permanently marked on the exterior of the assembly (see 6.2):

a. Manufacturer’s identification
b. Serial number
c. Military part number
d. Federal stock number
e. Manufacturer’s part number
f. Date of manufacture
g. Contract number
h. US.

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3.7 Workmanship. Workmanship shall be of the quality necessary to produce selector valves free from defects which may affect their serviceability or reduce, limit, or compromise performance (see 4.6.2).

3.8 Color. The color of exposed nonfunctional parts of the selector valve assembly shall be white (see 4.6.2).

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspections (see 4.3)

4.2 First article inspection. First article inspection shall be performed on two selector valve assemblies when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.4 (see table I) and the tests of table II.

TABLE I. Classification of defects.

Category	Defect	Method of inspection
Major:		
101	Dimensions affecting design or interchangeability not within tolerance (see 3.3 and 3.3.1).	SIE 1/
102	Material not compatible with hydraulic fluid per MIL-H-46170 (see 3.2.1).	Visual
103	Improper weight (see 3.3.2).	SIE 1/
Minor:		
201	Dimensions not affecting design or interchangeability not within tolerance (see 3.3 and 3.3.1).	SIE
202	Improper color (see 3.8).	Visual

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TABLE I. Classification of defects. (continued)

Category	Defect	Method of inspection
203	Improper marking (see 3.6).	Visual
204	Improper workmanship (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment

TABLE II. Classification of inspections.

Characteristic	Requirement	Inspection	First article	100% acceptance
Torque	3.4.1	4.6.4	X	X
Pressure drop	3.4.2	4.6.5	X	X
External leakage	3.4.3	4.6.6	X	X
Internal leakage	3.4.4	4.6.7	X	X
Proof pressure	3.4.5	4.6.8	X	X
Burst pressure	3.4.6	4.6.9	X	
High temperature	3.4.7.1	4.6.10.1	X	
Low temperature	3.4.7.2	4.6.10.2	X	
Shock resistance	3.4.7.3	4.6.10.3	X	X
Vibration resistance	3.4.7.4	4.6.10.4	X	
Endurance	3.4.8	4.6.11	X	

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.4 and the tests of 4.5.

4.4 Examination.

4.4.1 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ANSI/ANSI Z1.4. Each sample selected shall be inspected as specified herein for the defects listed in table I. Conformance to a lot shall be accepted when zero (0) defects are found and rejected when one (1) or more defects are found.

4.5 Acceptance tests (100 percent). Each selector valve assembly shall be subjected to the inspections specified in table II. Failure of any inspection shall be cause for rejection.

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4.6 Methods of inspection.

4.6.1 Materials. Conformance to 3.2 thru 3.2.2 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.2.1, 3.3, 3.3.1, 3.3.2, 3.6, 3.7, and 3.8 shall be determined by examination for the defects listed in table I. Examination shall be visual, tactile or by measurement with SIE.

4.6.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the following:

a. Ambient temperature: 70 ± 20°F (21 ± 11°C)
b. Hydraulic fluid temperature: 100 ± 10°F (38 ± 6°C)
c. Hydraulic fluid operating pressure: 3000 psig (20.6 MPa)

4.6.4 Torque test. To determine conformance to 3.4.1, 1000 psig (6.88 MPa) shall be applied to port 1 with port 2 blocked. A measurement of torque shall verify values of 8 ± 6 inch pounds (0.9 ± 0.68 N-m) for valve handle movement out of detent and not in excess of 15 inch pounds (1.7 N-m) for handle movement between detented positions.

4.6.5 Pressure drop test. To determine conformance to 3.4.2, with handle in open position, pressure is applied to one port to obtain a flow of 5 gpm (18.9 L/min) at the other port. A measurement of the pressure differential between ports shall verify the pressure drop shall not exceed 750 psi (5.16 MPa). At operational environmental extremes pressure drop shall be verified at 1500 psi (10.3 MPa) maximum.

4.6.6 External leakage test. To determine conformance to 3.4.3, visual examination of the assembly shall verify no external leakage during conformance verification.

4.6.7 Internal leakage test. To determine conformance to 3.4.4, with handle of valve in closed position, a pressure of 3000 psig (20.6 MPa) shall be applied to port 1 at a rate of 380 ± 20 psig (2.6 ± .14 MPa) per second for a 2-minute period. Internal leakage in port 2 shall be verified at 0.003 in^3 (0.05 cm^3) maximum for a 1-minute period. Internal leakage at port 2 shall be verified not to exceed 0.006 in^3 (0.1 cm^3) per minute subsequent to endurance test (see 4.6.11).

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4.6.8 Proof pressure test. To determine conformance to 3.4.5, with valve in closed position, a pressure of 4500 ± 50 psig (31 ± 0.35 MPa) shall be applied at a rate of 380 ± 20 psig (2.6 ± 0.14 MPa) per second for a 3-minute period. Selector valve shall be observed to verify absence of permanent deformation, rupture, or external leakage.

4.6.9 Burst pressure test. To determine conformance to 3.4.6 with valve enclosed position, a pressure of 7500 ± 50 psig (51.6 to 0.34 MPa) shall be applied at a rate of 380 ± 20 psig (2.6 ± 0.14 MPa) per second to the pressure port (port 1) for a 3-minute period. Selector valve shall be observed to verify absence of permanent deformation, rupture, or external leakage. The valve assembly is not expected to operate after this test.

4.6.10 Environmental testing. To determine conformance to 3.4.7, valve assembly shall be subjected to tests specified in 4.6.10.1 through 4.6.10.4.

4.6.10.1 High temperature test. Selector valve assembly shall undergo procedure II, operation method 501.3, MIL-STD-810. High temperature storage shall be 160°F (71°C) and maintained for 36 hours. At the conclusion of the storage temperature test, the assembly shall be stabilized at 125°F (51°C) and subjected to the 100 percent acceptance tests of table II. After testing, return the valve assembly to normal ambient temperature (see 4.6.3) and verify conformance to the 100 percent acceptance tests of table II.

4.6.10.2 Low temperature test. Selector valve assembly shall undergo procedure I, storage method 502.3, MIL-STD-810. Low temperature storage shall be -65°F (-54°C) and maintained for 12 hours. At the conclusion of storage temperature test, the assembly shall be stabilized at -25°F (-31°C) and subjected to the 100 percent acceptance tests of table II. After testing, return the valve assembly to normal ambient temperature (see 4.6.3) and verify conformance to the 100 percent acceptance tests of table II.

4.6.10.3 Shock resistance test. To determine conformance to 3.4.7.3, selector valve assembly shall be subjected to procedure I, functional shock method 516.4, MIL-STD-810. Configuration of pulse shall be half sine wave of 40 ± 4 g peak amplitude and 18 ± 3 ms duration measured at the 10 percent amplitude points.

4.6.10.4 Vibration resistance test. To determine conformance to 3.4.7.4, selector valve assembly shall be subjected to category 8-ground mobile, method 514.4, MIL-STD-810 with the following exceptions:

a. Vibration test profile shall be shown in figure 1 and conducted in 125°F (51°C) ambient temperature. Time schedule for vibration resistance test shall total 80 minutes per axis, including 13.33 minutes dwell time at each resonance.

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b. Connections and instrumentation shall be attached to the assembly to permit operation and testing as specified in (c) through (e).
c. A resonant frequency search shall be conducted along each axis at 70° ± 20°F (21° ± 11°C) over the frequency range of 5-500 Hz.
d. The valve assembly shall be placed in a temperature chamber in order to conduct the sinusoidal cycling and resonance/dwell test. Temperature, test levels, and time schedules shall be as specified in 4.6.10.1 and 4.6.10.2.
e. After vibration, the assembly shall be stabilized at 70° ± 20°F (21° ± 11°C) and subjected to the 100 percent acceptance tests of table II.

4.6.11 Endurance test. To determine conformance to 3.4.8, the selector valve assembly shall be cycled 10,000 times with 1,000 psig (6.88 MPa) applied to port 1 and port 2 blocked. The valve assembly, after cycling, shall be subjected to conformance tests in 4.6.1 through 4.6.5.

NOTE: A cycle shall consist of actuating the valve assembly from one detent to the other and then back again.

4.6.12 Internal cleanliness. To determine conformance to 3.5, the selector valve assembly shall be tested as follows:

a. Prior to any performance testing, establish the internal cleanliness of the assembly by providing an operating pressure of 3000 psig (20.6 MPa) and then establish a flow between the two ports. Maintain the flow until the effluent meets the cleanliness requirements as shown in table III. Refer to MIL-STD-1246, level 200 for additional information if needed.
b. At the conclusion of testing, re-establish the assembly's internal cleanliness by performing the same flushing and effluent check. Once valve internal cleanliness has been achieved, drain the effluent and seal the assembly with a noncontaminating, tamper, proof-seal, adequately tagged and if identified, before subjecting it to preparation for delivery or storage processing.

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TABLE III. Particle cleanliness levels.

Level	Particle size, micrometers (μm)	Count per 1 ft^2	Count per 0.1 m^2	Count per liter
200	15	4189	4520	41890
200	25	1240	1340	12400
200	50	170	184	1700
200	100	16	17.3	160
200	200	1.0	1.08	10

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Selector valves furnished under this specification are intended for dumping 900 psig (6.2 MPa) regulated pressure before introduction of ballistic computer corrections, and to drain the main accumulator.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DoDISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. When first article is required (see 3.1).
d. Identification and marking if other than as specified (see 3.6).
e. Inspection conditions if other than as specified (see 4.6.3).
f. Selection of applicable packaging requirements (see 5.1).

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6.3 Subject term (key word) listings.

Hydraulic fluid
Hydraulic pump
Selector valve port

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of changes.

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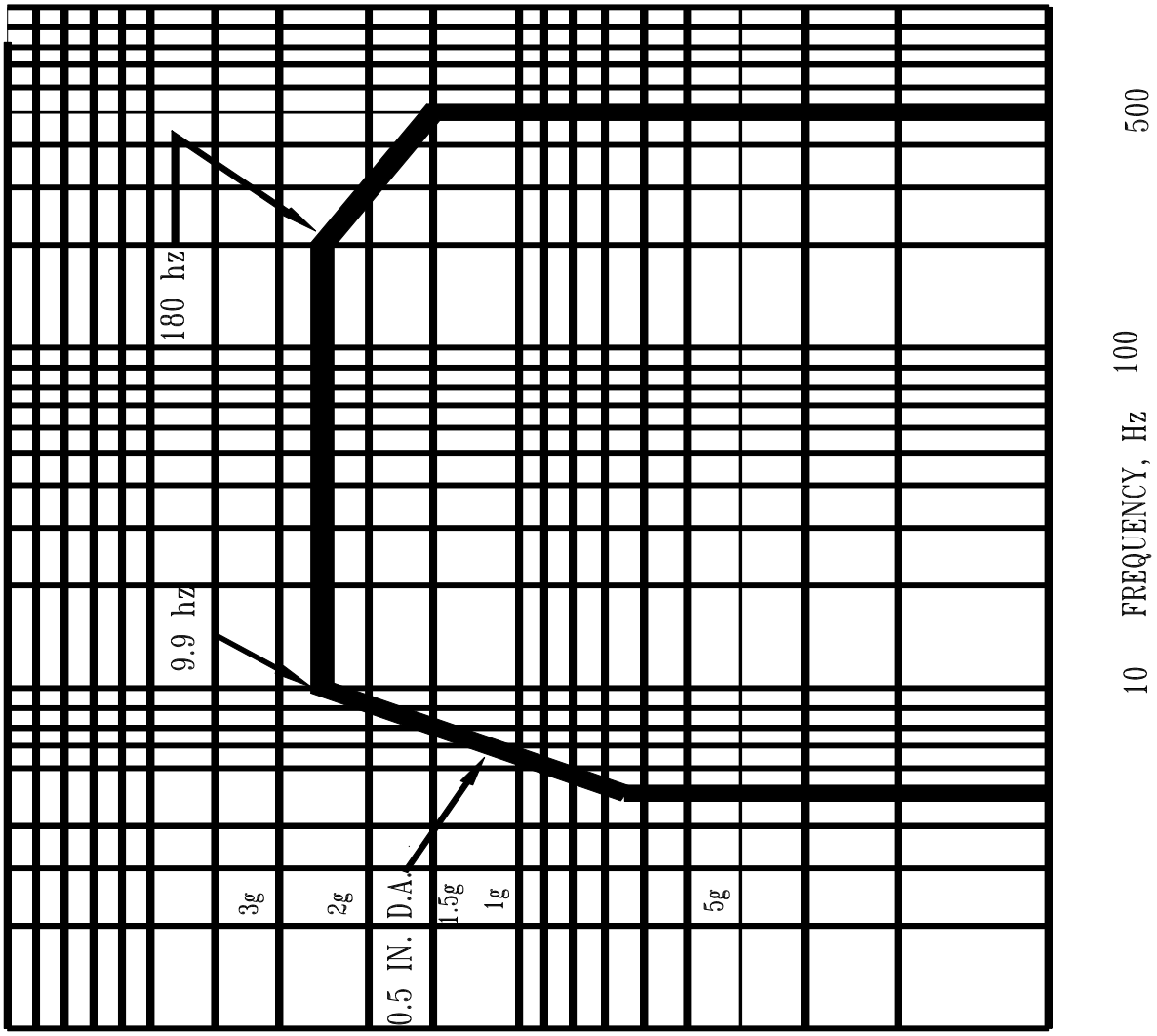


FIGURE 1. Vibration profile

Custodians:
Army - AT

Review activity:
DLA - CS

Preparing activity:
Army - AT

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